www.csvtuonline.com

Mech 329/6

Printed Pages – 8

337312 (37)

BE (3rd Semester)

Examination, April-May, 2013

Branch : Mechanical

NUMERICAL ANALYSIS & COMPUTER PROG. (C & C++)

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Part (a) of each question is compulsory. Solve other parts as per the internal choice given.

Q. 1. (a) The true value of a physical property is 10/3, but it is noted as 3.333. Give absolute error and relative error in noting down the value. **2**

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

Solve any two :

(b) Find a root of an equation $f(x) = 3x + \sin x - e^x = 0$ in the range [0, 1] using internal halving and regula falsi methods, compare the methods for three iterations only. **7**

(c) Solve the following system of equations using Gauss-Seidel method (correct to 2 decimal places) **7**

$$8x - 3y + 2z = 20$$

$$4x + 11y - z = 33$$

$$6x + 3y + 12z = 35$$

www.csvtuonline.com

www.csvtuonline.com

(d) Use the Newton-Raphson method to estimate the root of $f(x) = e^{-x} - x = 0$, employing an initial guess of $x_0 = 0$. State your answer after four iterations. 7

Q. 2. (a) For the data given below obtain value of y at x = 2 using linear interpolation : 2

x :	0	3
y :	0	2

Solve any one :

(b) Fit a second order polynomial to the data given in the table : 14

x_i :	0	1	2	3	4	5
y_i :	2.1	7.7	13.6	27.2	40.9	61.1

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

(c) The population of a certain town is give below. Find the rate of growth of population i 1931, 1941, 1961 and 1971. 14

Year :	1931	1941	1951	1961	1971
Population in thousands :	40.62	60.80	79.95	103.56	132.65

Q. 3. (a) State the condition for selecting number of sub-intervals for applying Simpson's Three-Eighth's Rule. 2

Solve any two :

(b) When a train is moving at 25 m/s, steam is shut off and brakes are applied. The speed of the train V in m/s and time t in seconds is given as : 7

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

t :	0	10	20	30	40	50	60
V :	25.0	18.1	12.3	7.6	4.0	1.6	0

Find the distance moved by train before coming to a halt using (i) Trapezoidal rule (ii) Simpson's one-third rule.

(c) Find the value of y(1.1), using Runge-Kutta method of fourth order given that : 7

$$\frac{dy}{dx} = y^2 + xy \; ; \; y(1) = 1$$

(d) Using Milne's method find y(4.4) given $5xy^1 + y^2 - 2 = 0$ given y(4) = 1, y(4.1) = 1.0049, y(4.2) = 1.0097 any y(4.3) = 1.0143. 7

www.csvtuonline.com

www.csvtuonline.com

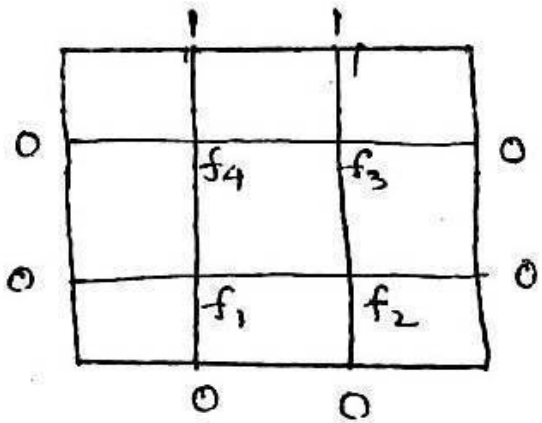
www.csvtuonline.com

Q. 4. (a) Classify the following equation as ellipti hyperbolic or parabolic.

$$xf_{xx} + yf_{yy} = 0, \; x > 0, \; y > 0$$

Solve any two :

(b) Solve the equation $f_{xx} + f_{yy} = 0$ in the doma



by Gauss-Seidel's method. (Only fi iterations)

www.csvtuonline.com

www.csvtuonline.com

(c) Solve the equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ subject to the conditions $u(x, 0) = \sin \pi x$, $0 \le x \le 1$, $u(0, t) = u(1, t) = 0$ using Crank-Nicolson method. Take $h = \frac{1}{3}$, $k = \frac{1}{36}$ carryout computation for two levels. 7

(d) Solve the poisson equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = -4(x+y)$ over the square mesh with sides $x = 0$, $y = 0$, $x = 4$ and $y = 4$ with $u = 0$ on the boundary, taking the mesh length into one unit. (Solve for 10 iterations). 7

Q. 5. (a) Write C expression for the following algebraic expression. 2

$$\text{Root} = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$$

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

Solve any two :

(b) Write a program in 'C' which can accept three integers as input and sort them in ascending order. 7

(c) Given the coordinates (x, y) of a center of a circle and its radius, write a program which will determine whether a point lies inside the circle, on the circle or outside the circle. 7

(d) Write a program to find the factorial value of any number entered through the keyboard. 7

www.csvtuonline.com